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Guideline: [PDF 202605 BMSE3153 Assignment Specification.pdf](#)

Practical 1

Detail → [Sem4 Master Doc](#)

1. Analyse the background of your selected software project.

1(a) project development team

a) Briefly describe the project development team (the project development team is your group) (hints: give your group a name, years of experience in the IT industry, group expertise, etc). Make sure 1(a) and 2(a) are different)

- Newly established technology startup named **Curuto Susake**
- Comprised of four co-founding software engineers from the same academic cohort.
- Collective experience of six years in the information technology industry.
- Core expertise includes:
 - Building secure full-stack web applications.
 - Managing database setups.
 - Creating responsive user interfaces.
- Utilizes the modern MERN technology stack: MongoDB, Express.js, React.js, and Node.js for high-performance system delivery.

1(b) Project Client Profile and Motivation

b) Briefly describe the project client, the client's nature of business, and the reasons the client wants to undertake the project.

- **Client Overview:** **LingJiaBan Co.**, a traditional HR and headhunting firm operating nationwide in Malaysia.
- **Business Role:** Acts as a corporate intermediary, connecting businesses with professional workers and aiding job seekers in securing employment.
- **Current Operations:** Operates offline using legacy workflows, including physical paper resumes, manual calls, and disorganized storage, leading to operational bottlenecks.
- **Issues:** Experiences lost applicant information, administrative delays, and a slow talent-matching process due to manual processes.
- **Project Goals:** Transition to a digital agency by creating a centralized, web-based Jobs Recruitment Website.
- **Expected Benefits:** Eliminate paperwork, automate keyword resume screening, and provide instant interview scheduling tools, reducing operational turnaround time by over 70%.
- **Market Positioning:** Aims to compete with digital platforms like JobStreet by expanding their services nationwide.

2. Perform project initialisation activities

2(a)[Window] Estimate the project costs

(This may include hardware, software, manpower, etc. The entire project development duration must be between 3 and 5 months.)

Duration : 4 months

Activity Cost Estimate (Project Cost Baseline)

WBS Items	Month 1	Month 2	Month 3	Month 4	Totals
1. Project Management					
Project Manager / Team Leader (Poong Foo Jing)	2,500	2,500	2,500	2,500	10,000
2. Software Development Team					
Lead Full-Stack Developer (Chong Ye Keh)	3,500	3,500	3,500	3,500	14,000
UI/UX Frontend Developer (Alvin Chong)	3,000	3,000	3,000	3,000	12,000
Database Admin & Risk Specialist (Chai Zhen Yi)	3,000	3,000	3,000	3,000	12,000
3. Hardware					
Developer Laptop (Existing/upgrade cost)	8,000	-	-	-	8,000
Testing Devices	1,500	-	-	-	1,500
Backup Storage Device	500	-	-	-	500
4. Software					
Domain Name	100	-	-	-	100
Cloud Hosting	300	300	300	300	1,200
Development Tools / Plugins	500	-	-	-	500
5. Testing & Deployment					
System Testing	-	-	1,000	1,000	2,000
User Acceptance Testing	-	-	-	1,000	1,000

Deployment & Documentation	-	-	500	500	1,000
6. Reserve	-	-	3,000	3,000	6,000
TOTAL	22,900	12,300	16,800	18,800	70,800
WBS Items	Month 1	Month 2	Month 3	Month 4	Totals
1. Project Management					
Project Manager / Team Leader (Poong Foo Jing)	2,500	2,500	2,500	2,500	10,000
2. Software Development Team					

Basis of Estimate (Assumptions)

Cost Item	Assumption
Project Manager	Responsible for client communication, sprint management, scheduling and deployment coordination
Lead Full-Stack Developer	Handles software architecture, backend/frontend integration and implementation planning
UI/UX Frontend Developer	Handles interface design, user experience and quality assessment planning
Database Admin & Risk Specialist	Handles database design, resume storage security, risk analysis and privacy compliance
Hardware	Includes development computers, testing devices and backup storage
Software	Uses mainly free development tools; cost only includes domain, hosting and required plugins
Reserve	Covers unexpected technical problems, additional services or project changes

Determine Budget

Project Cost = RM70,800

The approved budget required to complete the **CareerLink Recruitment Website Project within 4 months** is:

RM70,800

The budget includes:

- Project team manpower
- Hardware resources
- Software services
- Testing activities
- Deployment preparation
- Contingency reserve

2(b) **[KDE]** objective

In one sentence, state the **objective** of this project (must conform to the SMART criteria)

To develop and deploy a web-based Jobs Recruitment Website within 4 months that centralizes job postings, automates resume screening, and enables online interview scheduling, reducing recruitment processing time by at least 70% for LingJiaBan Co.
Specific: Develop a Jobs Recruitment Website with resume screening and interview scheduling.

Measurable: Reduce recruitment processing time by 70%.

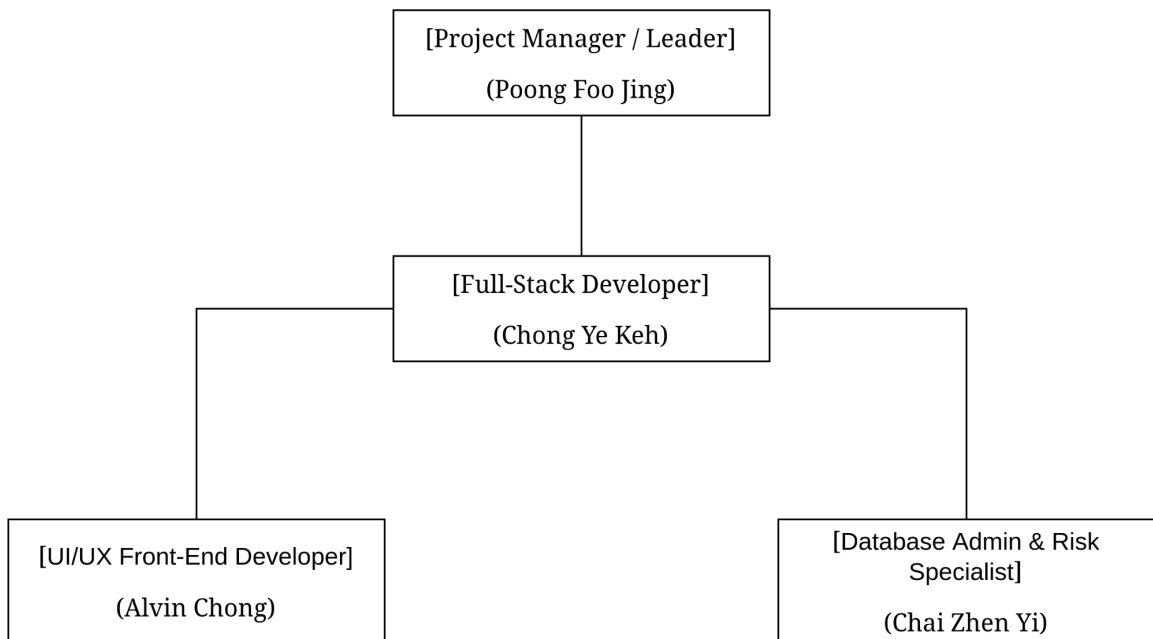
Achievable: Uses the team's MERN stack expertise.

Relevant: Solves the client's manual recruitment and data management issues.

Time-bound: Complete and deploy within 4 months.

2(c) **[Curry]** project team structure

Illustrate your **project team structure** graphically. Provide a list of team members' names, job positions, and roles.



Team Members, Positions, and Roles Directory

Team Member Name	Job Position	Project Management & Report Responsibilities
Poong Foo Jing	Project Manager / Team Leader	Responsible for managing client relations with LingJiaBan , coordinating team sprints, tracking deadlines, establishing the system communication framework, and handling overall project deployment logistics.
Chong Ye Keh	Lead Full-Stack Developer	Responsible for selecting the software architecture, mapping out the implementation schedule and milestones, and configuring the timeline in Microsoft Project to ensure development stages run smoothly.
Alvin Chong	UI/UX Frontend Developer	Responsible for designing the user interface layout for the job seeker portal, creating the 3-Level Work Breakdown Structure (WBS) for website features, and setting up the system's Quality Assessment testing plan.
Chai Zhen Yi	Database Admin & Risk Specialist	Responsible for structuring the database schema to handle secure resume storage, defining the platform's SMART operational goals,

		evaluating technical risks (like data breaches), and ensuring system data privacy compliance.
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2(d) [Curry] Stakeholders register

No	Stakeholder (Individual or Group)	Job Position (Individual Only)	Role (Client Side Only)	Expectation
1	LingJiaBan Management	Managing Director	Project Sponsor	Ensure the system is delivered on schedule, within budget, and achieves business objectives.
2	Recruitment Officers	Recruitment Officer	Primary System User	Reduce manual recruitment tasks and improve candidate search, screening, and interview coordination.
3	Employer Companies	-	Client Customer	Easily advertise vacancies, review applications, and monitor recruitment progress through the platform.
4	Job Seekers	-	End User	Quickly search jobs, submit applications, and track application status through one platform.
5	Curuto Susake Project Team	-	-	Receive clear requirements and successfully deliver a reliable system that meets client expectations.
6	System Administrator	HR System Administrator	System Administrator	Manage users, maintain system performance, and ensure data security and availability.

Note:

- must include the project team and client (e.g., project sponsor, potential users of the system, etc)
- You only need to fill in the role column, if the stakeholder is from the client side.

2(e) [Curry] Communication plan

No	Name of the Communication (Method)	Purpose	Frequency	Audience	Owner
1	Kick-off Meeting (Face-to-Face)	Introduce the project team, confirm project objectives, discuss scope, and establish project expectations.	Once (Project Start)	LingJiaBan Management, Curuto Susake Project Team	Project Manager
2	Weekly Team Stand-up (Google Meet)	Discuss project progress, review completed tasks, identify issues, and coordinate upcoming development activities.	Weekly	Curuto Susake Project Team	Project Manager
3	Project Status Report (Email)	Provide updates on project progress, completed deliverables, identified risks, and upcoming milestones.	Bi-weekly	LingJiaBan Management	Project Manager
4	Technical Review Meeting (Google Meet)	Review technical progress, discuss system integration issues, and ensure development activities remain aligned.	Weekly	Curuto Susake Project Team	Lead Full-Stack Developer

5	Requirements Verification Workshop (Microsoft Teams)	Validate system requirements, confirm functional features, and ensure client expectations are accurately understood.	Once (Planning Phase)	LingJiaBan Management, Recruitment Officers, Curuto Susake Project Team	Project Manager
6	Instant Messaging (WhatsApp)	Communicate urgent matters, share quick updates, and coordinate daily project-related activities efficiently.	As Needed	Curuto Susake Project Team, LingJiaBan Representative	Project Manager

2(f) [Hyprland] Project scope:

Illustrate graphically the scope of the system to be developed (show 3 levels: system, modules, and functions). Briefly describe the functionalities of each module.

1.0 LingJiaBan Recruitment Website
└ 1.1 User Management Module └ 1.1.1 User Registration └ 1.1.2 User Login & Logout └ 1.1.3 Password Recovery └ 1.1.4 Profile Management └ 1.1.5 Role-Based Access Control └ 1.2 Job Seeker Module └ 1.2.1 Create Job Seeker Profile └ 1.2.2 Upload Resume └ 1.2.3 Search Job Vacancies └ 1.2.4 Apply for Jobs └ 1.2.5 Track Application Status └ 1.3 Employer Module └ 1.3.1 Create Company Profile └ 1.3.2 Post Job Vacancy └ 1.3.3 Manage Job Postings └ 1.3.4 View Applicants └ 1.3.5 Shortlist Candidates └ 1.4 Recruitment Management Module └ 1.4.1 Search Candidate Database └ 1.4.2 Filter Candidates by Skills └ 1.4.3 Match Candidates to Jobs

1.4.4 Schedule Interviews 1.4.5 Update Recruitment Status
1.5 Resume Management Module 1.5.1 Resume Upload 1.5.2 Resume Storage 1.5.3 Resume Viewing 1.5.4 Resume Search 1.5.5 Resume Download
1.6 Notification Module 1.6.1 Application Notifications 1.6.2 Interview Notifications 1.6.3 Job Posting Notifications 1.6.4 Account Notifications 1.6.5 Email Alerts
1.7 Administration Module 1.7.1 Manage Users 1.7.2 Manage Employers 1.7.3 Manage Job Listings 1.7.4 Generate Reports 1.7.5 Monitor System Activities

Module	Description
1.1 User Management Module	Responsible for managing user accounts and authentication. This module allows users to register, log in, recover forgotten passwords, update personal information, and access system features according to their assigned roles (Job Seeker, Employer, or Admin).
1.2 Job Seeker Module	Provides functionality for job seekers to create and maintain professional profiles, upload resumes, search for available job opportunities, submit applications, and monitor the progress of their applications through the recruitment process.
1.3 Employer Module	Enables employers to manage their company profiles and recruitment activities. Employers can post job vacancies, edit or remove job postings, review applicants, and shortlist suitable candidates for further consideration.
1.4 Recruitment Management Module	Supports LingJiaBan recruiters and administrators in managing the recruitment process. This module allows staff to search candidate records, filter applicants based on skills or qualifications, match candidates with suitable job vacancies, schedule interviews, and track recruitment progress.
1.5 Resume Management Module	Handles the storage, organization, and retrieval of candidate resumes. It enables job seekers to upload resumes while allowing employers and recruiters to view, search, and download resume documents efficiently.

1.6 Notification Module	Facilitates communication between users by sending automated notifications and alerts. Users receive updates regarding job applications, interview schedules, account activities, and other important recruitment-related events through email or system notifications.
1.7 Administration Module	Provides administrative functions for managing the overall system. Administrators can manage user accounts, oversee employers and job listings, generate reports, monitor system activities, and ensure smooth operation of the recruitment platform.

2(g) [KDE] Exclusions:

List 2 items/services/jobs and/or hardware/software that will not be provided or covered in this project.

To score high → Provided an in depth answer that clearly met the details of the criteria.

1. Third-Party Background Verification Services (Service/Job Exclusion)

This project focuses on the technical architecture and functional development of the software system. Professional background verification is categorized as a specialized human resource and legal service that falls outside the defined Project Scope

Boundary Definition

-The system is designed to provide the tools for data entry and management, but it will not perform external investigative tasks such as criminal record checks, academic qualification authentication, or employment history verification

Specialism Constraints

-As software projects involve multiple specialisms (e.g., database and UI specialists), investigative legal services represent a "technical expertise" gap that the current project team is not staffed to provide

Operational Responsibility

-These verification activities require interaction with external investigative agencies, which is a Procurement activity that remains the responsibility of LingJiaBan Co. and the respective employers

Risk Mitigation

-By excluding these services, the project avoids the Risk of delays caused by third-party verification timelines, ensuring the project remains focused on its primary "mantra": delivering a quality system on time and within budget

2. Cloud Hosting and Domain Subscription (Hardware/Software Exclusion)

In project appraisal, it is vital to distinguish between Development Costs (one-time costs to build the system) and Operating Costs (ongoing costs to keep the system running)

Cost Categorization

-Cloud hosting fees, domain name registrations, and long-term server maintenance are classified as Operating Costs This project's budget and planning are strictly allocated to Development Costs, such as system design, coding, and testing

Life Cycle Termination

-The Project Life Cycle for this undertaking concludes at the Closing Process, which involves the formal handover of the product to the customer. Once the software is delivered and the project is terminated, the transition to a live environment (hosting) becomes an organizational operational task rather than a project task

Resource Planning

-Providing long-term maintenance would require a different Staffing model and a "maintenance project" structure that is not part of the current Project Management Plan (PMP)

Stakeholder Expectation Management

-Clearly defining this exclusion in the Project Charter ensures LingJiaBan Co. understands they must secure their own infrastructure for the software's "operating life" to avoid financial deviations later in the project

